

Financial Literacy Simulator

A Gamified Life Simulation for Learning Finance by Building Systems

January 27, 2026

The Problem We're Solving

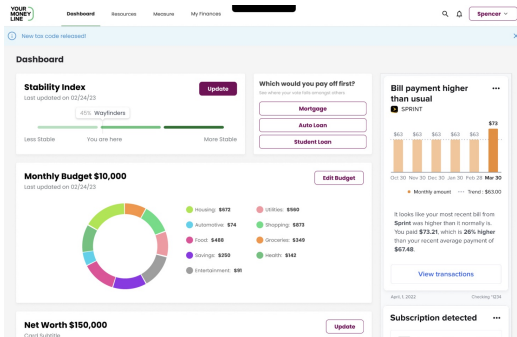
- Most young adults earn money before they understand money
- Financial mistakes compound quietly over years
- Traditional finance education is:
 - Static
 - Advice-heavy
 - Low on consequences
- Question: **What if people could experience their financial future early?**

Project Overview

Financial Literacy Simulator (FLS) is a gamified life-simulation app where:

- Users simulate their financial life from first salary
- Time is compressed: **1 week = 1 month of life**
- Every decision compounds over simulated decades

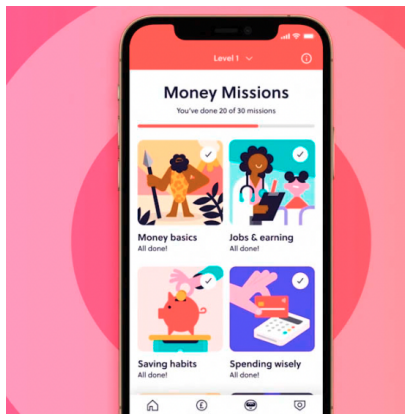
Built by students as a learning-by-building engineering project.



Target Users

- Young adults (18–30) with low financial literacy
- Students experimenting with life decisions
- Building real-world simulations

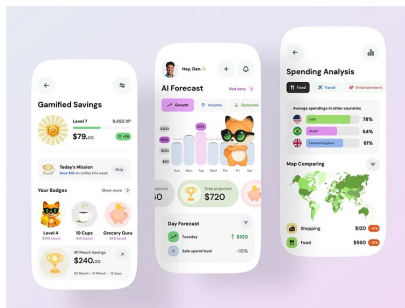
Key idea: Learn finance not by advice, but by consequence.



Learning Outcomes for Players

By playing, users learn to:

- Understand tradeoffs: spending vs saving vs investing
- Experience compounding and volatility
- Appreciate emergency funds and insurance
- See links between health, career, and money
- Accept that there is no single “correct” strategy



Learning Outcomes for students

By building the system, students learn to:

- Translate ambiguous ideas into requirements
- Design stateful, time-based systems
- Model real-world domains in software
- Build clean APIs and data models
- Follow engineering best practices (Git, testing, reviews)

Game Concept & Lifecycle

Game Start

- First salary credit

Game End

- Retirement age reached
- OR bankruptcy

Every run creates a unique financial story.

Time Model

- 1 game week = 1 simulated month
- Each week includes:
 - ① Salary credit
 - ② Mandatory expenses
 - ③ Player decisions
 - ④ Random events
 - ⑤ State updates
 - ⑥ Weekly summary

Core Player Metrics

Each player maintains:

- Net Worth
- Cash balance
- Investments
- Debt
- Career level
- Well-being score

These metrics evolve every simulated month.

Net Worth

Net Worth = Assets – Liabilities

Influenced by:

- Savings
- Investments
- Physical assets (house, etc.)
- Debt (loans, EMIs, credit)

Net worth tells the long term story, not just cash.

Random Life Events

Examples:

- Medical emergency
- Job loss or promotion
- Market crash or boom
- Family obligations

Purpose:

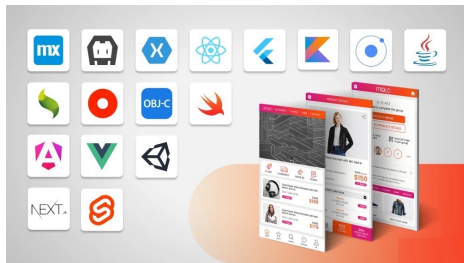
- Model uncertainty
- Reward resilience, not luck

Why This Project Matters

- Real-world domain complexity
- Clear state transitions
- Natural need for clean architecture
- Easy to extend incrementally
- Visible impact of engineering decisions

Technology Stack

- **Mobile App:** Flutter (Dart) / React
- **Backend:** Node.js with Express.js
- **Database:** Amazon DynamoDB
- **Cloud & Services:** AWS (IAM, API Gateway, CloudWatch)
- **Local Development:** LocalStack for AWS service simulation
- **Version Control:** Git & GitHub (PRs, reviews)
- **Design & UX:** Figma (wireframes and mockups)
- **Testing:** Manual testing, API testing, basic automation



People don't fail at money because they're bad at math.

They fail because they never see consequences early enough.

This simulator lets them live decades in minutes.